

Entropie

Relative Entropie von endlichen Quantenzuständen

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Abstract

This is the abstract.

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1 Section 1

1.1 Some Definitions

Question 1.1. Definieren hier die f-Divergenz allgemein für positive lineare Funktionale über von Neumann Algebren. Haben wir dadurch später Vorteile, oder willst du das einfach erstmal so allgemein wie möglich machen? Für die Physik und viele Resultate werden wir schätze ich mal Zustände verwenden.

Definition 1.2. Let $\varphi, \psi : \mathcal{M} \rightarrow \mathbb{C}$ be positive linear functionals over a von Neumann algebra $\mathcal{M}$ such that $\varphi \ll \psi$ was das nochmal genau??? and $f : (0, \infty) \rightarrow \mathbb{R}$ be a convex function with $f(1) = 0$ (brauche ich das hier???). The *f-divergence* of φ and ψ is given by

$$\begin{aligned}
 D_f(\varphi \mid \psi) &= \int_1^\infty f''(\gamma) E_\gamma(\varphi \mid \psi) + \gamma^{-3} f''(\gamma^{-1}) E_\gamma(\psi \mid \varphi) d\gamma \\
 &= \int_1^\infty f''(\gamma) E_\gamma(\varphi \mid \psi) + \int_0^1 \gamma f''(\gamma) E_{\frac{1}{\gamma}}(\psi \mid \varphi) d\gamma.
 \end{aligned} \tag{1}$$

Definition 1.3. Let φ and ψ be as above. (sicher? oder doch Zustände?) The hockeystick divergence $E_\gamma(\varphi \mid \psi)$ is defined as follows:

$$E_\gamma(\varphi \mid \psi) := (\psi - \gamma\varphi)_+(\mathbb{1}) \tag{2}$$

Some facts about the hockeystick divergence.

Proposition 1.4. Let φ_1, φ_2 be positive linear functionals over a von Neumann algebra, and p_1, p_2 are (pos. lin. functionals or states?)

$$\lim_{t \rightarrow \infty} E_t(\varphi_1 \mid \varphi_2) = \varphi_1(p_2^\perp). \tag{3}$$

$$\lim_{t \rightarrow \infty} E_t(\varphi_1, \varphi_2) = 0 \iff p_1 \leq p_2. \tag{4}$$

1.2 First Inequality

In [SV16] there are a lot of inequalities for f-divergences in the context of probability measures.

We want to generalize them for states over arbitrary von Neumann algebras (even vNA of type III).

We start with inequality (84) from [SV16].

Proposition 1.5. Let $f, g : (0, \infty) \rightarrow \mathbb{R}$ convex functions with $f(1) = g(1)$, $f'(1) = g'(1)$ and $f(t) \leq g(t)$ for all $t \in (0, \infty)$. Then, the following inequality holds:

$$D_f(\varphi \mid \psi) \leq D_g(\varphi \mid \psi). \quad (5)$$

φ and ψ are states over an arbitrary von Neumann algebra.

Remark. Alexandrov's theorem states that every convex function defined on an open domain is twice differentiable almost everywhere.

Remark. By assumption $f'(1) = g'(1)$ indirectly require, that f and g are differentiable at $x = 1$.

2 Section 2

2.1 Subsection A

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References

- [SV16] Igal Sason and Sergio Verdu. “ f -Divergence Inequalities”. In: *IEEE Transactions on Information Theory* 62.11 (Nov. 2016), pp. 5973–6006. ISSN: 1557-9654. DOI: 10.1109/tit.2016.2603151. URL: <http://dx.doi.org/10.1109/TIT.2016.2603151>.